Relational Schema (PK underlined, *FK italic*)

USER(userID, username, email, passwordHash, subscriptionTier, dateJoined)

ARTIST(artistID, name, country, primaryGenre, bio)

ALBUM(albumID, title, releaseDate, albumType, coverImageURL, *artistID*)

SONG(songID, title, duration, trackNumber, isExplicit, *albumID*)

PLAYLIST(playlistID, name, description, visibility, createdDate, *userID*)

LISTEN_HISTORY(historyID, timestamp, playDuration, *userID*, *songID*)

USER_ARTIST(*userID*, *artistID*, followedDate)

SONG_ARTIST(*songID*, *artistID*, role)

PLAYLIST_SONG(*playlistID*, *songID*, position, addedDate)

LISTENING_SNAPSHOT(snapshotID, snapshotMonth, totalMinutes, mostActiveDay, *userID*)

BCNF PROOF:

USER(userID, username, email, passwordHash, subscriptionTier, dateJoined)

Key: userID
FDs: userID → username, email, passwordHash, subscriptionTier, dateJoined
BCNF: LHS is a key ⇒ BCNF.

ARTIST(artistID, name, country, primaryGenre, bio)

Key: artistID
FDs: artistID → name, country, primaryGenre, bio
BCNF: LHS is a key ⇒ BCNF.

ALBUM(albumID, title, releaseDate, albumType, coverImageURL, artistID)

Key: albumID
FDs: albumID → title, releaseDate, albumType, coverImageURL, artistID
BCNF: LHS is a key ⇒ BCNF.

SONG(songID, title, duration, trackNumber, isExplicit, albumID)

Key: songID
FDs: songID → title, duration, trackNumber, isExplicit, albumID
BCNF: LHS is a key ⇒ BCNF.
(Optional extra constraint if you want: (albumID, trackNumber) → songID, title, … if track numbers are unique per album. Only include if you enforced it.)

PLAYLIST(playlistID, name, description, visibility, createdDate, userID)

Key: playlistID
FDs: playlistID → name, description, visibility, createdDate, userID
BCNF: LHS is a key ⇒ BCNF.

LISTEN_HISTORY(historyID, timestamp, playDuration, userID, songID)

Key: historyID
FDs: historyID → timestamp, playDuration, userID, songID
BCNF: LHS is a key ⇒ BCNF.
(If you instead used (userID, songID, timestamp) as key, list that.)

USER_ARTIST(userID, artistID, followedDate)

Key: (userID, artistID)
FDs: (userID, artistID) → followedDate
BCNF: LHS is a key ⇒ BCNF.

SONG_ARTIST(songID, artistID, role)

Key: (songID, artistID)
FDs: (songID, artistID) → role
BCNF: LHS is a key ⇒ BCNF.

PLAYLIST_SONG(playlistID, songID, position, addedDate)

Key: (playlistID, songID) *(or (playlistID, position) if you enforce unique positions)*
FDs: (playlistID, songID) → position, addedDate
BCNF: LHS is a key ⇒ BCNF.

LISTENING_SNAPSHOT(snapshotID, snapshotMonth, totalMinutes, mostActiveDay, userID)

Key: snapshotID
FDs: snapshotID → snapshotMonth, totalMinutes, mostActiveDay, userID
BCNF: LHS is a key ⇒ BCNF.